# Remote Controller with Fail-Safe Update

Concept of Operations

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|---|---|
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# 1.0 Introduction

## 1.1 Project Description

### 1.1.1 Background

**Purpose:** The System lets an operator on the ground command, monitor, and update a controller in space that nobody can reach, and keeps that controller working between contacts.

**Need:** A controller that nobody can reach must stay operable, observable, and correctable for its whole life. Operation, observability, and maintenance including in-flight modification over the specified lifetime are three areas a space software specification must cover [3, Annex B, items 5.11–5.13, p. 113].

**Worst-Case Failure:** A failed update can leave the controller unable to run any firmware and unable to accept new firmware. Without physical access, that state is permanent.

**Rationale:** The loss has happened. NASA records a mission terminated after an in-flight software upload omitted a patch for a defective countdown timer; its recommendations include test-bed verification of in-flight software updates [4, Lesson 0559, p. 199].

### 1.1.2 Assumptions and Constraints

**C-1** The ground cannot talk to the controller directly. Every command, report, and firmware image passes through the onboard computer.

**C-2** All software is built to NASA's software engineering requirements. Rigor per part is set by its software class [5, Sec. 3.5.1, p. 24; App. C].

**C-3** One engineer builds and checks the System. No independent team reviews the work.

**C-4** The System runs on commercial development boards, not on spacecraft hardware. It can show how the System behaves, not that it would survive launch or space.

**A-1** Once launched, physical access to the onboard computer or controller is impossible; all fixes and updates must arrive by radio.

**A-2** The controller can lose power at any moment, even halfway through writing new firmware. The System must survive that without losing the controller [2, App. A, Table 2, p. 63].

**A-3** The ground can reach the spacecraft only during line-of-sight contact windows **[TBD-5]**. Between contacts the System must work on its own.

**A-4** One person operates the ground station. No second person checks a command before it is sent, so the System itself must refuse wrong commands.

**A-5** Radiation in space can flip single bits in memory, so stored firmware can change on its own. The System must not run firmware that has changed [2, App. A, Table 2, p. 63].

***Note.*** *A constraint (C) is a limit the System must work within; it is imposed, not chosen. An assumption (A) is a condition taken as true without proof; the design depends on it, and if it proves false the design may be wrong.* ***[TBR-**n**]*** *marks a proposal the author has not yet accepted;* ***[TBD-**n**]*** *marks a value not yet known. Both are listed in App. C.*

## 1.2 Overview of the Envisioned System

### 1.2.1 Overview

The System comprises two spatially separated operational segments. The **ground segment** on Earth includes the operator, the mission operations center running ground software, and the ground station antenna [6, Sec. 11.2, p. 305]. The **space segment** in orbit consists of the onboard computer and the controller, which operate without physical access.

Communication between segments is strictly established via a radio link during line-of-sight contact windows [**TBD-5**]. Outside contact windows, the space segment operates autonomously. Within the spacecraft, the onboard computer communicates with the controller over the data link during normal operation. If the controller's main firmware fails to execute, the onboard computer utilizes the reprogramming link to restore functional firmware independently of the controller's primary software. Figure 1 illustrates this segment layout.

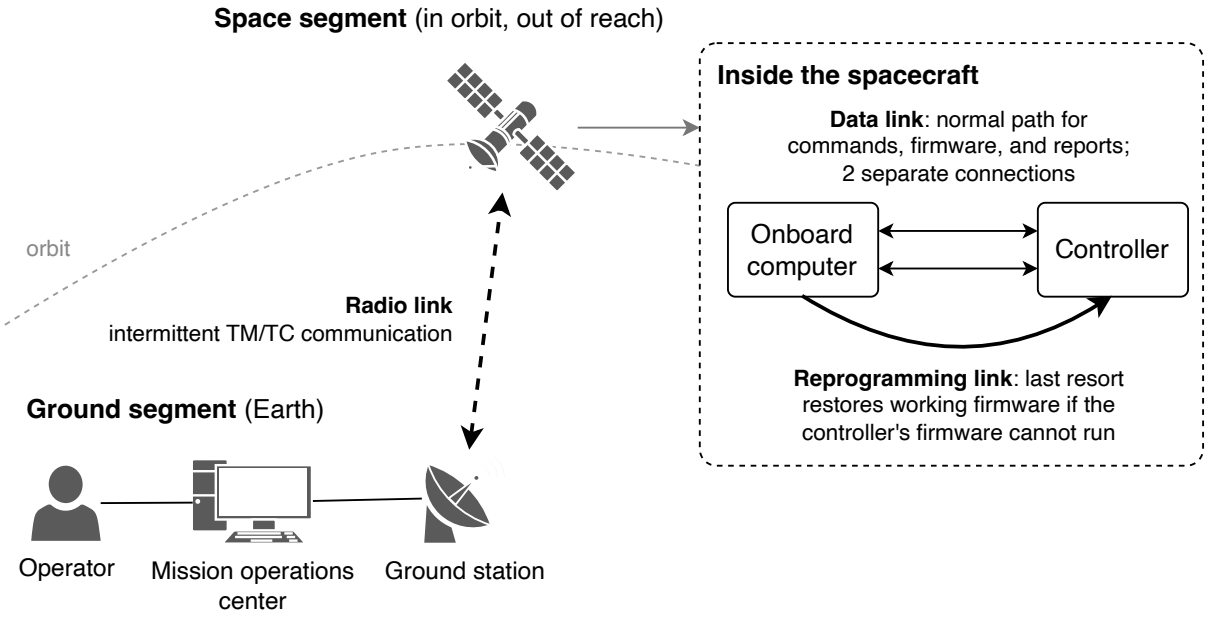



Figure 1: Ground and space segments, joined only by the radio link during contact windows. Inside the spacecraft, the data link handles normal communication, while the reprogramming link restores working firmware if the controller's firmware cannot run.

### 1.2.2 System Scope

The system context defines the boundary of the System: the system of interest, its users, and the external elements that interface with it [7, Sec. 8.3, p. 24].

| Context Element | System Boundary |
|---|---|
| System of Interest | Onboard computer, controller, ground software in the MOC |
| Users | Ground operator |
| External Elements | Radio equipment, spacecraft power supply |

- **Enabling systems:** firmware build tools and bench test setups required for development and verification, external to operational scope.
- **In scope:** commanding, telemetry, firmware updates, and autonomous fault handling between contact windows [**TBR-5**].
- **Out of scope:** onboard computer software updates [**TBR-7**], and the radio equipment, which is used as given.

# 2.0 Documents

## 2.1 Applicable Documents

[1] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016. This outline follows its App. S, pp. 251–253, tailored per its Sec. 3.11.1, p. 34: Section 7.0 not applicable; App. C added for open items per its App. C, pp. 197–198.

## 2.2 Reference Documents

[2] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022. Off-nominal scenarios cite App. A, Table 2.
[3] ESA, *Space engineering — Software*, ECSS-E-ST-40C Rev. 1, 30 April 2025.
[4] NASA, *Fault Management Handbook*, NASA-HDBK-1002, Draft 2, 2 April 2012. A draft, not a released standard.
[5] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[6] NASA, *State-of-the-Art of Small Spacecraft Technology*, Chapter 11, "Ground Data Systems and Mission Operations", 2026 edition. Ground system architecture in Sec. 11.2 and Fig. 11.2, pp. 305–306.
[7] NASA, *NASA Systems Modeling Handbook for Systems Engineering*, NASA-HDBK-1009A, 12 March 2025.

# 3.0 Description of Envisioned System

## 3.1 Needs, Goals and Objectives

**NEED** A controller that nobody can reach must stay operable, observable and correctable for its whole life. The three properties are three areas a space software specification must cover: operation, observability, and maintenance including in-flight modification over the specified lifetime [3, Annex B, items 5.11–5.13, p. 113].

**G-5** Operable: the controller keeps running its function and acts on the operator's commands. Capabilities CMD, FM.

**G-6** Observable: the ground can know what the System is doing, how healthy it is, and what happened between contacts. Capabilities TM.

**G-7** Correctable: the controller's firmware can be replaced for its whole life without losing the controller. Capabilities UPD.

**G-1** No update, and no failure during one, leaves the controller without firmware it can run.

**G-2** The System acts only on firmware and commands from the operator.

**G-3** Losing one connection of the data link stops nothing.

**G-4** Between contacts the System handles faults alone and reports them later **[TBR-5]**.

**O-1** Every scenario in Section 6.0 is demonstrated on hardware and recorded.

**N-1** Retired during drafting; now G-7. NASA requires a single need statement [1, Sec. 4.1, p. 49].

**N-2** Retired during drafting; now G-6.

**N-3** Retired during drafting; now G-5.

## 3.2 Overview of System and Key Elements

**Operator** Releases firmware, decides commands, reads reports. Works through the mission operations center.

**Mission operations center** Runs the ground software: commands the spacecraft and monitors its performance [6, Sec. 11.2, p. 305]. Packages released firmware, checks commands before they are sent, and stores reports.

**Ground station** The antenna and radio that carry the radio link during contacts [6, Sec. 11.2, p. 305].

**Onboard computer** Checks and passes on firmware and commands, stores reports, watches and restores the controller.

**Controller** Runs its function, reports, keeps known-good firmware to fall back on.

### 3.3 Interfaces

**Radio link** The only connection between the ground and the spacecraft. Commands and firmware go up; reports come down. It exists only during contact windows.

**Data link** The normal path between the onboard computer and the controller. Commands and new firmware go to the controller; reports come back. It works only while the controller's firmware runs. It has two separate connections: if one fails, the other carries everything.

**Reprogramming link** The last-resort path, used only when the controller's firmware cannot run. It restores working firmware without any help from the controller's own software.

### 3.4 Modes of Operations

**OPERATE** Runs confirmed firmware. New firmware can arrive [**TBR-6**].

**TRIAL** Runs new firmware not yet judged good [**TBD-6**].

**FALLBACK** Runs previous or last-resort firmware after a failed trial or start.

**RECOVERY** Runs no firmware. The onboard computer restores it through the reprogramming link.

### 3.5 Proposed Capabilities

**TM-1** Send regular health reports: alive, firmware, uptime, last restart reason, temperature [**TBR-4**].

**TM-2** Report important events at once [**TBR-4**].

**TM-3** Timestamp every report [**TBR-4**].

**TM-4** Store reports between contacts, send all at the next one, report any lost.

**TM-5** Report the health of each data-link connection [**TBR-4**].

**CMD-1** Accept commands only from the operator. Refuse replayed commands.

**CMD-2** Answer every command: received, refused with reason, or done with result.

**CMD-3** Refuse commands not allowed in the current state.

**CMD-4** Run a resent command at most once.

**CMD-5** Dangerous commands need two steps: arm, then execute.

**CMD-6** Refuse impossible values.

**CMD-7** Let the operator cancel an update, leaving known-good firmware running.

**CMD-8** Run commands at a set time. Refuse expired commands.

**CMD-9** Log every command and its answer.

**FM-1** Notice a silent controller and try to bring it back [**TBR-5**].

**FM-2** Carry on over the other connection when one dies [**TBR-5**].

**FM-3** Restart crashed onboard programs and continue [**TBR-5**].

**FM-4** Record every fault and action for the ground [**TBR-5**].

**UPD-1** Take firmware from the ground through dropping links.
**UPD-2** Refuse firmware that is damaged, not from the operator, for something else, or below the minimum version.
**UPD-3** Get the firmware to the controller.
**UPD-4** Switch the controller to it [**TBD-3**].
**UPD-5** Decide whether it is good [**TBD-4**].
**UPD-6** Go back to working firmware if not, with nobody's help.
**UPD-7** Tell the ground what happened.
**UPD-8** Never leave the controller without runnable firmware.
**VER-1** Degrade the radio link on purpose (test only).
**VER-2** Cut controller power and break one connection on command (test only) [**TBR-10**].

## 4.0 Physical Environment

Follows from the situation in which nobody can reach the System. Response legend [1, App. S, p. 252]: OPERATE = all functions continue; SURVIVE = functions stop and return when the condition ends.

| Condition | Response | Scenario |
|---|---|---|
| No ground contact | OPERATE | OPS-110 |
| Controller power loss at any instant | SURVIVE | OPS-340 |
| Single-bit memory error | OPERATE | OPS-400 |
| Temperature range | [**TBD-9**] | — |

## 5.0 Support Environment

Firmware is built and released on the ground and sent up during contacts, one transfer across several if needed. The onboard computer keeps the last known-good firmware. Update rate [**TBD-8**]. Nothing is repaired after deployment.

## 6.0 Operational Scenarios

### 6.1 Nominal Conditions

**OPS-100** Update: release, upload over one or more contacts, check, transfer [**TBR-6**], switch [**TBD-3**], judged good [**TBD-4**], confirmed, outcome reported.
**OPS-110** Telemetry: reports stamped and stored between contacts, all sent at the next one.
**OPS-120** Scheduled command: received during a contact, runs at its time, answer sent at the next contact.

### 6.2 Off-Nominal Conditions

Trigger, then expected outcome. Sources in brackets are items of [2, App. A, Table 2].
**OPS-200** Ground link lost mid-transfer: resumes next contact where it stopped.
**OPS-210** Command not from operator or replayed: refused, logged [2, p. 62].
**OPS-220** Answer lost, command resent: not run again, "already done".
**OPS-230** Command wrong for current state: refused with reason [2, p. 63].
**OPS-240** Impossible value: refused with reason [2, p. 62].
**OPS-250** Dangerous command not armed: refused with reason.
**OPS-260** Command expired: refused with reason [2, p. 63].
**OPS-270** Report storage full: ground told how many were lost.
**OPS-300** Firmware damaged in transfer: refused before the controller [2, p. 63].

**OPS-310** Firmware not from operator: refused.
**OPS-320** Firmware below minimum version: refused.
**OPS-330** Firmware for another controller: refused [2, p. 63].
**OPS-340** Power lost during firmware write: previous firmware runs; retry possible [2, p. 63].
**OPS-350** New firmware hangs: back to previous, unaided.
**OPS-360** New firmware restart loop: back to previous after **[TBD-7]** starts [2, p. 64].
**OPS-370** New firmware never judged good: back to previous.
**OPS-380** Previous firmware also bad: last-resort firmware runs.
**OPS-390** Operator cancels update: known-good firmware left running.
**OPS-400** Bit error in stored firmware: detected, not run **[TBR-9]** [2, p. 63].
**OPS-410** No firmware can run: restored through the reprogramming link [2, p. 63].
**OPS-500** One data-link connection fails: everything continues on the other; reported.
**OPS-510** Both data-link connections fail: controller keeps working; loss reported; reprogramming link still usable **[TBR-8]**.
**OPS-520** Controller silent: noticed, recovery tried, recorded **[TBR-5]**.
**OPS-530** Onboard program crash: it restarts and resumes any update; the controller is unaffected **[TBR-5]**.

## 7.0 Impact Considerations

Not applicable: bench demonstration.

## 8.0 Risks and Potential Issues

**R-1** Both data-link connections may share software; one defect could cut both.
**R-2** Controller memory may not hold last-resort plus two working firmwares.
**R-3** Second data-link connection and reprogramming link not yet shown on bench.
**R-4** Demonstration date **[TBD-2]**; undemonstrated scenarios are declared, not dropped.

## A Acronyms

CMD command; ConOps Concept of Operations; FM fault management; MOC mission operations center; OPS operational scenario; TC telecommand; TBD To Be Determined; TBR To Be Resolved; TM telemetry; UPD update; VER verification.

## B Glossary

**Confirmed** Firmware judged good after its trial; only confirmed firmware is known-good.
**Last-resort** Known-good firmware that updates never replace (golden image).
**Switch** The controller stops running old firmware and starts the new.
**Arm** First step of a dangerous command; changes nothing alone.
**Expiry** Time after which a waiting command is refused.

## C Open Items

Unmarked statements were stated or accepted by the author on 9–10 October 2026. TBR items are resolved before the hazard analysis.

### Resolved

**TBR-1** Spacecraft controller, ground contacts only, no physical access, bench stand-in, bit errors occur, no repair. Accepted by the author, 10 October 2026.

**TBR-2** Controller power can drop at any instant, including mid-write. Accepted by the author, 10 October 2026.

**TBR-3** One operator. Accepted by the author, 10 October 2026.

### Open

**TBR-4** TM-1, TM-2, TM-3, TM-5 as written (shown 10 Oct, not yet accepted).

**TBR-5** G-4, FM-1 to FM-4 as written (shown 10 Oct, not yet accepted).

**TBR-6** Controller keeps working while new firmware arrives.

**TBR-7** Onboard computer's own update out of scope.

**TBR-8** Reprogramming link still usable with both data-link connections lost.

**TBR-9** Stored firmware checked before start; damaged copy skipped.

**TBR-10** Test setup can cut power and break one connection.

**TBD-1** Project name and document reference.

**TBD-2** Demonstration date.

**TBD-3** Switch automatically on arrival, or only on operator command.

**TBD-4** Who judges new firmware good: controller or operator.

**TBD-5** Contact window length and gap (sourced, or a test parameter).

**TBD-6** Trial time limit (requirements).

**TBD-7** Failed starts before going back (requirements).

**TBD-8** Update rate.

**TBD-9** Temperature range.